

AP Precalculus: 2026–2027

Unit 3: Trigonometric and Polar Functions Lesson 3.7: Sinusoidal Function Context and Data Modeling (2–3 instructional periods, 55 minutes each)

Primary Objective: Students will construct sinusoidal function models of periodic phenomena by extracting amplitude, vertical shift, period, and phase shift from contextual data, and will use those models to predict values within a contextual domain. (*Big Idea: Functions; Skills 1.C, 3.C*)

Priority Ideas & Skills

AP Skills

1.C Construct new functions — Build a sinusoidal model from a data set by estimating key parameter values or by performing a sinusoidal regression with technology.

3.C Support conclusions with data — Evaluate the reasonableness of a sinusoidal model by comparing predicted values to actual data, and clarify the contextual domain over which the model is useful.

Key Understandings (EKs)

EK 3.7.A.1: The smallest input-value interval between consecutive maxima (or consecutive minima) in the data estimates the period.

EK 3.7.A.2: Amplitude = $\frac{\max - \min}{2}$; vertical shift/midline = $\frac{\max + \min}{2}$.

EK 3.7.A.3: A known input-output pair is compared to the model to estimate the phase shift.

EK 3.7.A.4: Technology can be used to perform sinusoidal regression and to estimate parameters from data.

EK 3.7.A.5: Sinusoidal models are valid only over a contextual domain and are used to predict output values from input values.

Vocabulary, Concepts, & Theorems

Term

Definition / Note

sinusoidal model

A function of the form $f(x) = A \sin(B(x - C)) + D$ (or cosine form) used to represent a periodic real-world quantity.

amplitude

Half the range of output values: $A = \frac{\max - \min}{2}$. Measures how far the function deviates from its midline.

vertical shift (midline)

$D = \frac{\max + \min}{2}$; the horizontal axis about which the sinusoidal function oscillates.

period

The input-value distance between consecutive maxima (or minima) in the data; equals $\frac{2\pi}{B}$ in the model.

phase shift

C ; determined by comparing a known data point (e.g., a maximum) to the standard sine/cosine maximum location.

contextual domain

The restricted set of input values over which the model is meaningful given the real-world situation.

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8–10 minutes):

- **Reading sinusoidal graphs (Lessons 3.5–3.6):** Identify amplitude, period, and midline from a graph — connects directly to extracting the same quantities from data tables.
- **Computing $(\max + \min)/2$ and $(\max - \min)/2$ from a data set:** A prerequisite numerical skill targeted directly in Warm-Up Problem 2.
- **Function transformation parameters A, B, C, D :** Students should recall how each parameter shifts or scales the parent sin or cos function before applying those ideas to real data.

Hook — Entry Event

Scenario: Display a scatter plot of average monthly high temperature ($^{\circ}\text{F}$) for a U.S. city over one calendar year. The data trace a smooth wave: low in January, peaking in July, dropping again by December.

Discussion prompts:

1. “What kind of function could model this shape? How do you know?”
2. “What information from the table would help you write the equation?”
3. “If we use this model to predict the temperature in month 15, should we trust that prediction? Why or why not?”

Key tension to surface: Students know sinusoidal functions from Lessons 3.5–3.6 but have only worked with equations given directly. Today they must *extract* parameters from messy real-world data — a new challenge.

Transition: “Today we’ll learn a systematic four-step process for turning any periodic data set into a sinusoidal model, then evaluate how well our model fits.”

Lesson — Instruction Sequence

Step 1 — Estimate Period from Data (8 min)

Present a 12-month temperature data table. Ask: “Where are the consecutive maxima?” Model how to identify the x -values of two consecutive peaks and compute their difference (EK 3.7.A.1). Then compute $B = \frac{2\pi}{k}$. Verify by checking consecutive minima; both should give the same period.

Step 3 — Determine Phase Shift (10 min)

For the sine model $f(x) = A\sin(B(x - C)) + D$: standard sine reaches its maximum at $B(x - C) = \frac{\pi}{2}$, i.e., $x = C + \frac{\pi}{2B}$. Set this equal to the x -value of the data’s first maximum; solve for C (EK 3.7.A.3). Validate: substitute a second data point to check closeness. Tip: if a maximum occurs at $x = x_0$, using a cosine model gives $C = x_0$ directly.

Step 5 — Technology: Sinusoidal Regression (8 min)

Demonstrate entering data in Desmos or a graphing calculator and running a sinusoidal regression (EK 3.7.A.4). Compare regression parameters to hand estimates. Discuss why they may differ: the regression minimizes total error across all points, not just the extremes.

Step 2 — Compute Amplitude and Midline (8 min)

Show the formulas $A = \frac{\max - \min}{2}$ and $D = \frac{\max + \min}{2}$ (EK 3.7.A.2). Apply to the temperature data. Discuss the midline in context: “the average between the hottest and coolest month.” Emphasize that amplitude is measured *from* the midline, not from zero.

Step 4 — Validate and Predict (10 min)

Evaluate the model at several data points. Discuss residuals (predicted minus actual). Use the model to predict a value (e.g., temperature in month 14) and state the contextual domain (EK 3.7.A.5). Caution: far outside the domain, real-world factors (climate change, relocation) may break the assumption of perfect periodicity.

Step 6 — Contextual Domain Discussion (5 min)

Ask: “Should we trust this model to predict temperatures 50 years from now?” Reinforce EK 3.7.A.5: validity requires the periodic assumption to hold. Real climate data, population trends, or equipment changes may violate that assumption outside the observed domain.

Pacing note: Steps 1–3 in Period 1; Steps 4–6 + activity in Periods 2–3.

Active Monitoring

- **Period vs. half-period:** Students measure from a maximum to the next minimum (half-period). Prompt: “We need two peaks — maximum to maximum, not maximum to minimum.”
- **Amplitude vs. maximum value:** Students write $A = \max$. Prompt: “The amplitude is measured from the midline. What is the midline here?”
- **Phase shift sign errors:** The form $\sin(B(x - C))$ shifts right by C (positive C means shift right). Students flip the sign. Reinforce: substitute $x = C$ — do you get a maximum output?

- **Contextual domain over-extension:** Students apply the model beyond the data range without comment. Ask: “*What real-world constraint limits meaningful input values for this model?*”

Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity. Assign tiers or allow structured self-selection with teacher guidance.

- **Tier R (Reinforce):** Given a complete data table with max, min, and the period already labeled, students compute A , D , and B using scaffolded formula templates and fill in the model $f(x) = A \sin(B(x - C)) + D$. A step-by-step guide and calculator are provided.
- **Tier A (Apply):** Given a tidal data table over 24 hours, students identify max, min, period, amplitude, midline, and phase shift; write the full model; and use it to predict the tide height at a specified time.
- **Tier E (Extend):** Given temperature data for two cities, students build models for both, compare parameters (which city is warmer on average? which has greater seasonal swing?), compute residuals, and determine which city’s climate is more faithfully modeled by a sinusoidal function.

Individual Work & Assessment

Exit ticket administered independently (final 5 minutes of Period 2 or 3).

1. A small 6-value data table is provided. Students identify the maximum, minimum, amplitude, and midline.
2. Students identify the period from consecutive maxima in the table and compute B .
3. Students use the given model to predict one output value and state whether that input is inside the contextual domain.

Data use: Sort exit tickets by which parameter caused the most errors. Use the most common error to open Lesson 3.8 with a 60-second targeted recap.

Reinforcement & Extension

Homework (Lesson 3.7): 5–6 problems covering: extracting parameters from a data table, writing a sinusoidal model, using the model to predict, evaluating model fit, and 2 AP-style multiple-choice items. An optional extension asks students to run a sinusoidal regression in Desmos and compare the regression equation to their hand-built model.

Preview of Lesson 3.8 — The Tangent Function: Ask students: “*Sine and cosine model bounded, wave-like phenomena. What if a periodic quantity shoots off toward infinity rather than staying bounded? What function might capture that behavior?*” Lesson 3.8 introduces the tangent function, its vertical asymptotes, and its definition as $\tan \theta = \sin \theta / \cos \theta$.